

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMIC DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 04-10-2024	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
September 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1305	664	78
	4.2 < ws < 7.6	768	1016	562
	ws > 7.6	108	469	3199
Number of points 8169 ; PC=67.6%; EBD=2.3%; POD1=59.8%; POD2=47.3%; POD3=83.3%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2015	140	0
	4.2 < ws < 7.6	166	1897	150
	ws > 7.6	0	112	3689
Number of points 8169 ; PC=93.0%; EBD=0.0%; POD1=92.4%; POD2=88.3%; POD3=96.1%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
Model data	ws < 12.0	6611	113	0
	12.0 < ws < 18.0	75	1086	38
	ws > 18.0	0	47	199
Number of points 8169 ; PC=96.7%; EBD=0.0%; POD1=98.9%; POD2=87.2%; POD3=84.0%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction		Observations			
degree		North	East	South	West
Model data	North	4754	215	0	64
	East	183	263	36	0
	South	28	83	675	83
	West	222	44	7	100
Number of points 6757 ; PC=85.7%; EBD=1.1%; POD1=91.7%; POD2=43.5%; POD3=94.0%; POD4=40.5%					

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction		Observations			
degree		North	East	South	West
Model data	North	5131	44	0	15
	East	44	548	23	0
	South	0	13	690	5
	West	12	0	5	227
Number of points 6757 ; PC=97.6%; EBD=0.0%; POD1=98.9%; POD2=90.6%; POD3=96.1%; POD4=91.9%					

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3011	882	9
	7.1 < rh < 15.3	1149	1419	224
	rh > 15.3	105	257	838
Number of points 7894 ; PC=66.7%; EBD=1.4%; POD1=70.6%; POD2=55.5%; POD3=78.2%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4084	153	0
	7.1 < rh < 15.3	181	2376	29
	rh > 15.3	0	29	1042
Number of points 7894 ; PC=95.0%; EBD=0.0%; POD1=95.8%; POD2=92.9%; POD3=97.3%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	7853	6	0
	50.0 < rh < 70.0	2	33	0
	rh > 70.0	0	0	0
Number of points 7894 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=84.6%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	4752	781	5
	1.9 < pwv < 3.5	185	1330	180
	pwv > 3.5	0	41	793
Number of points 8067 ; PC=85.2%; EBD=0.1%; POD1=96.3%; POD2=61.8%; POD3=81.1%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	4878	67	0
	1.9 < pwv < 3.5	59	2070	17
	pwv > 3.5	0	15	961
Number of points 8067 ; PC=98.0%; EBD=0.0%; POD1=98.8%; POD2=96.2%; POD3=98.3%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	537	554	217
	0.6 < see < 0.8	401	490	615
	see > 0.8	32	221	652
Number of points 3719 ; PC=45.1%; EBD=6.7%; POD1=55.4%; POD2=38.7%; POD3=43.9%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	966	171	4
	0.6 < see < 0.8	106	1052	117
	see > 0.8	1	132	1538
Number of points 3719 ; PC=87.0%; EBD=0.1%; POD1=90.0%; POD2=77.6%; POD3=92.7%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	942	0	148
	0.6 < see < 0.8	0	1228	148
	see > 0.8	28	37	1336
Number of points 3719 ; PC=94.3%; EBD=4.7%; POD1=97.1%; POD2=97.1%; POD3=81.9%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1072	0	4
	0.6 < see < 0.8	0	1350	4
	see > 0.8	1	5	1655
Number of points 3719 ; PC=99.8%; EBD=0.1%; POD1=99.9%; POD2=99.6%; POD3=99.5%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	956	564	41
	$3.4 < \tau < 5.8$	256	637	170
	$\tau > 5.8$	19	195	358
Number of points 3196 ; PC=61.0%; EBD=1.9%; POD1=77.7%; POD2=45.6%; POD3=62.9%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1167	112	1
	$3.4 < \tau < 5.8$	64	1211	67
	$\tau > 5.8$	0	73	501
Number of points 3196 ; PC=90.1%; EBD=0.0%; POD1=94.8%; POD2=86.7%; POD3=88.0%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1181	134	124
	$3.4 < \tau < 5.8$	31	1192	124
	$\tau > 5.8$	19	70	445
Number of points 3196 ; PC=88.2%; EBD=4.5%; POD1=95.9%; POD2=85.4%; POD3=64.2%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1231	0	6
	$3.4 < \tau < 5.8$	0	1390	6
	$\tau > 5.8$	0	6	563
Number of points 3196 ; PC=99.6%; EBD=0.2%; POD1=100.0%; POD2=99.6%; POD3=97.9%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	454	561
	0.6 < glf < 0.7	351	479
	glf > 0.7	68	288
Number of points 3322 ; PC=36.9%; EBD=10.2%; POD1=52.0%; POD2=36.1%; POD3=26.0%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	728	159
	0.6 < glf < 0.7	143	996
	glf > 0.7	2	173
Number of points 3322 ; PC=80.4%; EBD=0.2%; POD1=83.4%; POD2=75.0%; POD3=84.5%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	797	110
	0.6 < glf < 0.7	8	1200
	glf > 0.7	68	18
Number of points 3322 ; PC=84.6%; EBD=11.3%; POD1=91.3%; POD2=90.4%; POD3=58.0%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	868	9
	0.6 < glf < 0.7	3	1318
	glf > 0.7	2	1
Number of points 3322 ; PC=99.5%; EBD=0.2%; POD1=99.4%; POD2=99.2%; POD3=99.5%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1594	484
	4.6 < ws < 7.4	1031	1147
	ws > 7.4	77	306
Number of points 8165 ; PC=70.8%; EBD=2.0%; POD1=59.0%; POD2=59.2%; POD3=86.2%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2411	166
	4.6 < ws < 7.4	291	1671
	ws > 7.4	0	100
Number of points 8165 ; PC=92.0%; EBD=0.0%; POD1=89.2%; POD2=86.3%; POD3=97.2%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	6139	139
	12.0 < ws < 18.0	87	1361
	ws > 18.0	0	39
Number of points 8165 ; PC=96.1%; EBD=0.0%; POD1=98.6%; POD2=88.4%; POD3=86.2%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	4030	241	0		77
	East	40	298	41		0
	South	25	15	441		178
	West	743	17	20		936
Number of points 7102 ; PC=80.3%; EBD=0.6%; POD1=83.3%; POD2=52.2%; POD3=87.8%; POD4=78.6%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	4710	36	2		74
	East	36	529	7		0
	South	0	4	481		11
	West	92	2	12		1106
Number of points 7102 ; PC=96.1%; EBD=0.1%; POD1=97.4%; POD2=92.6%; POD3=95.8%; POD4=92.9%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3762	775	8
	8.9 < rh < 17.9	968	1430	286
	rh > 17.9	26	149	708
Number of points 8112 ; PC=72.7%; EBD=0.4%; POD1=79.1%; POD2=60.7%; POD3=70.7%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4622	137	0
	8.9 < rh < 17.9	134	2195	23
	rh > 17.9	0	22	979
Number of points 8112 ; PC=96.1%; EBD=0.0%; POD1=97.2%; POD2=93.2%; POD3=97.7%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	8056	4	0
	50.0 < rh < 70.0	3	49	0
	rh > 70.0	0	0	0
Number of points 8112 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=92.5%; POD3=NaN%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	4857	504	0
	2.1 < pwv < 3.7	266	1280	344
	pwv > 3.7	0	58	820
Number of points 8129 ; PC=85.6%; EBD=0.0%; POD1=94.8%; POD2=69.5%; POD3=70.4%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5064	42	0
	2.1 < pwv < 3.7	59	1784	22
	pwv > 3.7	0	16	1142
Number of points 8129 ; PC=98.3%; EBD=0.0%; POD1=98.8%; POD2=96.9%; POD3=98.1%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1369	706	76
	4.2 < ws < 7.6	650	948	563
	ws > 7.6	167	495	3200
Number of points 8174 ; PC=67.5%; EBD=3.0%; POD1=62.6%; POD2=44.1%; POD3=83.4%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4800	236	16
	East	174	283	70
	South	28	57	619
	West	185	29	13

Number of points 6757 ; PC=86.4%; EBD=1.1%; POD1=92.5%; POD2=46.8%; POD3=86.2%; POD4=55.5%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	2912	844
	7.1 < rh < 15.3	1251	1478
	rh > 15.3	105	236

Number of points 7897 ; PC=66.7%; EBD=1.3%; POD1=68.2%; POD2=57.8%; POD3=81.9%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	4671	818
	1.9 < pwv < 3.5	266	1293
	pwv > 3.5	0	41

Number of points 8067 ; PC=81.8%; EBD=0.4%; POD1=94.6%; POD2=60.1%; POD3=64.5%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	696	613	283
	0.6 < see < 0.8	554	824	817
	see > 0.8	90	343	1303
Number of points		5523 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1270	0	180
	0.6 < see < 0.8	0	1678	180
	see > 0.8	70	102	2223
Number of points		5523 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1827	873	91
	$3.4 < \tau < 5.8$	305	740	299
	$\tau > 5.8$	37	246	433
Number of points 4851 ; PC=61.8%; EBD=2.6%; POD1=84.2%; POD2=39.8%; POD3=52.6%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2076	188	233
	$3.4 < \tau < 5.8$	56	1559	233
	$\tau > 5.8$	37	112	590
Number of points 4851 ; PC=87.1%; EBD=5.6%; POD1=95.7%; POD2=83.9%; POD3=55.9%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	595	741
	0.6 < glf < 0.7	726	971
	glf > 0.7	105	294
Number of points 5040 ; PC=40.6%; EBD=10.2%; POD1=41.7%; POD2=48.4%; POD3=30.0%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1295	194
	0.6 < glf < 0.7	26	1776
	glf > 0.7	105	36
Number of points 5040 ; PC=83.8%; EBD=11.1%; POD1=90.8%; POD2=88.5%; POD3=57.1%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1605	502	89
	4.6 < ws < 7.4	1028	1142	399
	ws > 7.4	78	293	3038
Number of points 8174 ; PC=70.8%; EBD=2.0%; POD1=59.2%; POD2=59.0%; POD3=86.2%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4077	304	0
	East	42	229	30
	South	36	33	440
	West	683	5	32

Number of points 7102 ; PC=79.6%; EBD=0.6%; POD1=84.3%; POD2=40.1%; POD3=87.6%; POD4=76.2%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3551	724
	8.9 < rh < 17.9	1169	1451
	rh > 17.9	36	179

Number of points 8112 ; PC=69.8%; EBD=0.7%; POD1=74.7%; POD2=61.6%; POD3=66.0%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	4806	579
	2.1 < pwv < 3.7	317	1196
	pwv > 3.7	0	67

Number of points 8129 ; PC=83.3%; EBD=0.1%; POD1=93.8%; POD2=64.9%; POD3=65.8%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1051	646	107
	4.2 < ws < 7.6	879	863	553
	ws > 7.6	229	600	3179
Number of points 8107 ; PC=62.8%; EBD=4.1%; POD1=48.7%; POD2=40.9%; POD3=82.8%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4624	236	16
	East	284	250	95
	South	94	94	598
	West	128	15	9

Number of points 6690 ; PC=82.8%; EBD=1.9%; POD1=90.1%; POD2=42.0%; POD3=83.3%; POD4=26.3%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	2765	851
	7.1 < rh < 15.3	1355	1485
	rh > 15.3	116	211

Number of points 7854 ; PC=64.7%; EBD=1.8%; POD1=65.3%; POD2=58.3%; POD3=77.8%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	4660	862
	1.9 < pwv < 3.5	223	1188
	pwv > 3.5	0	89

Number of points 8000 ; PC=79.1%; EBD=0.5%; POD1=95.4%; POD2=55.5%; POD3=48.9%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	672	524	262
	0.6 < see < 0.8	509	757	784
	see > 0.8	145	458	1357
Number of points		5468 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1204	0	158
	0.6 < see < 0.8	0	1648	158
	see > 0.8	122	91	2245
Number of points		5468 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1751	814	42
	$3.4 < \tau < 5.8$	390	766	359
	$\tau > 5.8$	28	235	414
Number of points 4799 ; PC=61.1%; EBD=1.5%; POD1=80.7%; POD2=42.2%; POD3=50.8%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2042	167	240
	$3.4 < \tau < 5.8$	99	1516	240
	$\tau > 5.8$	28	132	575
Number of points 4799 ; PC=86.1%; EBD=5.6%; POD1=94.1%; POD2=83.5%; POD3=54.5%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	580	698	352
	0.6 < glf < 0.7	711	945	725
	glf > 0.7	119	330	528
Number of points 4988 ; PC=41.2%; EBD=9.4%; POD1=41.1%; POD2=47.9%; POD3=32.9%				

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1260	133	385
	0.6 < glf < 0.7	31	1810	342
	glf > 0.7	119	30	1220
Number of points 4988 ; PC=86.0%; EBD=10.1%; POD1=89.4%; POD2=91.7%; POD3=62.7%				

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	1442	444	102
	4.6 < ws < 7.4	1122	1095	381
	ws > 7.4	147	398	3043
Number of points 8174 ; PC=68.3%; EBD=3.0%; POD1=53.2%; POD2=56.5%; POD3=86.3%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	4006	317	8
	East	32	214	28
	South	80	33	427
	West	720	7	39

Number of points 7102 ; PC=76.9%; EBD=1.3%; POD1=82.8%; POD2=37.5%; POD3=85.1%; POD4=68.3%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3421	639
	8.9 < rh < 17.9	1241	1475
	rh > 17.9	94	240

Number of points 8112 ; PC=68.2%; EBD=1.2%; POD1=71.9%; POD2=62.7%; POD3=63.3%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	4727	714
	2.1 < pwv < 3.7	396	1054
	pwv > 3.7	0	74

Number of points 8129 ; PC=81.1%; EBD=0.1%; POD1=92.3%; POD2=57.2%; POD3=69.7%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration